Grade: Check Plus

Problem Set Assignment 2

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(1) Mean = Expected Value

$E[x] = \text{Sum } x P(x)$

$20 \times 0.05 = 1.00$
$21 \times 0.06 = 1.26$
$22 \times 0.07 = 1.54$
$23 \times 0.08 = 1.84$
$24 \times 0.09 = 2.16$
$25 \times 0.12 = 3.00$
$26 \times 0.15 = 3.90$
$27 \times 0.13 = 3.51$
$28 \times 0.10 = 2.80$
$29 \times 0.10 = 2.90$
$30 \times 0.05 = 1.50$

$E[x] = 1 + 1.26 + 1.54 + 1.84 + 2.16 + 3.00 + 3.90 + 3.51 + 2.80 + 2.90 + 1.50 = 25.41$ ✓Q1

↳ Mean Stock price = $\underline{\$25.41}$

$Var = E[x^2] - (E[x])^2$

$E[x^2] = \underline{653.17}$

$20^2 \times 0.05 = 400 \times 0.05 = 20$
$21^2 \times 0.06 = 441 \times 0.06 = 26.46$
$22^2 \times 0.07 = 484 \times 0.07 = 33.88$
$23^2 \times 0.08 = 529 \times 0.08 = 42.32$
$24^2 \times 0.09 = 576 \times 0.09 = 51.84$
$25^2 \times 0.12 = 625 \times 0.12 = 75$
$26^2 \times 0.15 = 676 \times 0.15 = 101.4$
$27^2 \times 0.13 = 729 \times 0.13 = 94.7$
$28^2 \times 0.10 = 784 \times 0.10 = 78.4$
$29^2 \times 0.10 = 841 \times 0.10 = 84.1$
$30^2 \times 0.05 = 900 \times 0.05 = 45$

Variance:

$Var = 653.17 - (25.41)^2$

$(25.41)^2 = 653.17$

$645.67 = 653.17$

$Var = 653.17 - 645.67 = \underline{\underline{7.50}}$

Standard Deviation:

$\sigma = \sqrt{7.50} = \underline{\underline{2.74}}$

(2a) 25.41 ± 2.74

$[22.67, 28.15]$

↳ Values within range = 23, 24, 25, 26, 27, 28

↳ $0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10 = \underline{\underline{0.67}}$ ✓ Q2a

(2b) $25.41 \pm 2(2.74) \rightarrow 25.41 \pm 5.48$

$[19.93, 30.89]$ → All possible values fall within range $= \underline{\underline{1.00}}$ ✓ Q2b

(3) Distribution of Option Value

Price	Option Value
20-25 → 0	0.05 + 0.06 + 0.07 + 0.08 + 0.09 + 0.12 = 0.47
26 → 1	0.15
27 → 2	0.13
28 → 3	0.10
29 → 4	0.10
30 → 5	0.05

✓ Q3

(4) Option = positive value

0.15 + 0.13 + 0.10 + 0.10 + 0.05 = 0.53

✓ Q4

(5) Mean = Expected Value

$E[x] = \text{Sum} \times P(x)$

$0 \times 0.47 = 0$

$1 \times 0.15 = 0.15$

$2 \times 0.13 = 0.26$

$3 \times 0.10 = 0.30$

$4 \times 0.10 = 0.40$

$5 \times 0.05 = 0.25$

$E[x] = 1.36$

Variance:

$E[x^2]$

$1^2 \times 0.15 = 0.15$

$2^2 \times 0.13 = 4 \times 0.13 = 0.52$

$3^2 \times 0.10 = 9 \times 0.10 = 0.90$

$4^2 \times 0.10 = 16 \times 0.10 = 1.60$

$5^2 \times 0.05 = 25 \times 0.05 = 1.25$

$E[x^2] = 4.42 \rightarrow$ Variance $= 4.42 - (1.36)^2$

$= 4.42 - 1.85$

$= 2.57$

Standard Deviation $= \sqrt{2.57} = \underline{\underline{1.60}}$